Lower heating
value
MJ/kg Hydrogen % m/m
Boiling point °C Nitrogen % m/m
Density at boiling
point

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SOURCE: MEPC.402(83).pdf
Sulphur % m/m
Gas fuel characteristics
Fuel type:
Fuel properties Fuel elemental analysis
Methane number EN16726:2015 Carbon % m/m
Lower heating
value
MJ/kg Hydrogen % m/m
Boiling point °C Nitrogen % m/m
Density at boiling
point
kg/m³ Oxygen % m/m
Pressure at
boiling point
Bar (abs) Sulphur % m/m
Methane, CH4 mol%
Ethane, C2H6 mol%
Propane, C3H8 mol%
Isobutane, i C4H10 mol%
N-Butane, n C4H10 mol%
Pentane, C5H12 mol%
C6+ mol%
CO2 mol%
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
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Gaseous emissions data
Mode 1 2 3 4 5 6 7 8 9 10
Power/Torque (%)
Speed (%)
Time at beginning of mode
Gaseous emissions data:
Sampling position
HC concentration (ppmC)
CH4 concentration (ppmC)*
recorded#
CH4 concentration (ppmC)* -
corrected#
N2O concentration (ppm)*
CO concentration (ppm)
CO2 concentration (%)
O2 concentration (%)

CH4 mass flow (kg/h)*#
N2O mass flow (kg/h)*
CO mass flow (kg/h)
CO2 mass flow (kg/h)
O2 mass flow (kg/h)
CH4 (g/kg)*#

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SOURCE: MEPC.402(83).pdf
corrected#
N2O concentration (ppm)*
CO concentration (ppm)
CO2 concentration (%)
O2 concentration (%)
CH4 mass flow (kg/h)*#
N2O mass flow (kg/h)*
CO mass flow (kg/h)
CO2 mass flow (kg/h)
O2 mass flow (kg/h)
CH4 (g/kg)*#
CH4 (g/kWh)* #
N2O (g/kg)*
N2O (g/kWh)*
* As applicable.
# As applicable to either liquid or gas fuel.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
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Engine parameters to be measured and recorded
Mode 1 2 3 4 5 6 7 8 9 10
Power/Torque (%)
Speed (%)
Time at beginning of mode
Engine data
Speed (rpm)
Power (kW)
Mean effective pressure (kPa)
Fuel rack/gas admission duration**
(mm/sec)
Liquid-to-gas fuel ratio (on mass basis)
Liquid Fuel flow (kg/h or m3/h*)
Gas Fuel flow (kg/h)
Exhaust flow (qmew) (kg/h)
Exhaust temperature at the sampling
point (oC)
Charge air coolant temperature in (oC)
Charge air coolant temperature out (oC)
Charge air temperature (oC)
Charge air reference temperature (oC)
Charge air pressure (kPa)

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SOURCE: MEPC.402(83).pdf
Exhaust temperature at the sampling
point (oC)
Charge air coolant temperature in (oC)
Charge air coolant temperature out (oC)
Charge air temperature (oC)
Charge air reference temperature (oC)
Charge air pressure (kPa)
Fuel-liquid temperature before the
engine (oC)
Fuel-gas temperature before the engine
(oC)
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
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Ambient data
Atmospheric pressure (kPa)
Intake air temperature (oC)
Intake air humidity (g/kg)
Relative humidity (RH) of intake air*
%
Air temperature at RH sensor* (oC)
Dry bulb temperature of intake air*
(oC)
Wet bulb temperature of intake air*
(oC)
* As applicable.
** Only for engines to be tested with gas fuel.
Abatement device:
The report should state whether reported data before or after device- hence the gaseous emission data page will need to be repeated, if both are
to be given.

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SOURCE: MEPC.402(83).pdf
* As applicable.
** Only for engines to be tested with gas fuel.
Abatement device:
The report should state whether reported data before or after device- hence the gaseous emission data page will need to be repeated, if both are
to be given.
Additionally, if both before and after data given, the analyser data should, if relevant, be repeated to cover all analysers used.
For each Mode Point, the following device data should additionally be recorded: Settings, Operating values and Consumption (specified rates).
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
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APPENDIX 3
FORM OF STATEMENT OF EMISSION VALUES FOR METHANE (CH4) AND/OR NITROUS OXIDE (N2O)
Issued in accordance with the Guidelines for test-bed and onboard measurements of methane (CH4) and/or nitrous oxide (N2O) emissions from marine diesel engines (resolution MEPC.402(83)) under the authority of the Government of:

(full designation of the country)

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SOURCE: MEPC.402(83).pdf
methane (CH4) and/or nitrous oxide (N2O) emissions from marine diesel engines (resolution MEPC.402(83)) under the authority of the Government of:

(full designation of the country)
by.....
(full designation of the competent person or organization authorized)
Particulars of applicant
Name of applicant
THIS IS TO DECLARE THAT:
1 the applicant has submitted to this Administration the information recommended by the Guidelines for test-bed and onboard measurements of CH4 and/or N2O emissions from marine diesel engines (resolution MEPC.402(83));
2 the emission value(s) have been established in accordance with the Guidelines for test-bed and onboard measurements of CH4 and/or N2O emissions from marine diesel